

18-799: Applied Computer Vision

Spring 2026

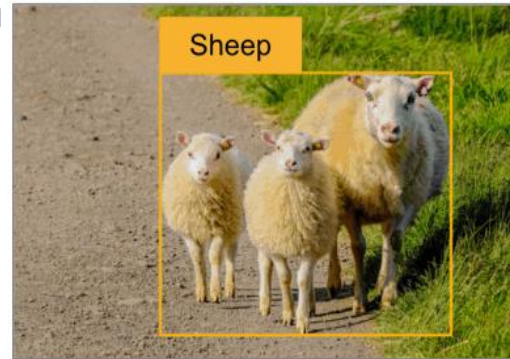
Introduction to Segmentation

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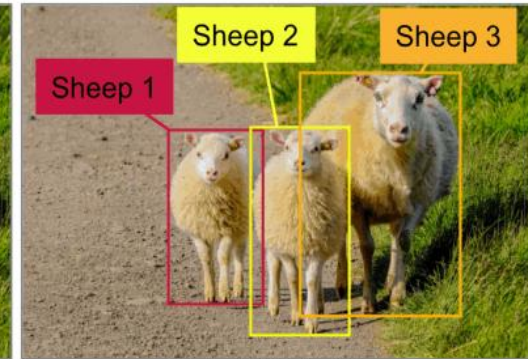
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What is Segmentation

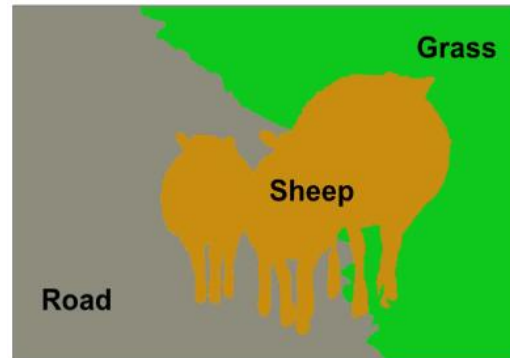
- In computer vision, segmentation refers to dividing an image into meaningful regions based on pixel similarity.
- **Semantic Segmentation** – Classifies each pixel into a specific class (e.g., sky, road, car).
- **Instance Segmentation** – Distinguishes different objects of the same category (e.g., multiple cars in an image).
- **Panoptic Segmentation** – Combines both semantic and instance segmentation.



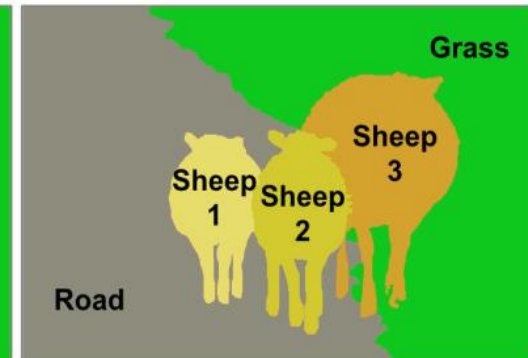
Classification + Localization



Object Detection



Semantic Segmentation



Instance Segmentation

Source: https://www.oreilly.com/ideas/introducing-capsule-networks?cmp=tw-data-na-article-ainy18_thea

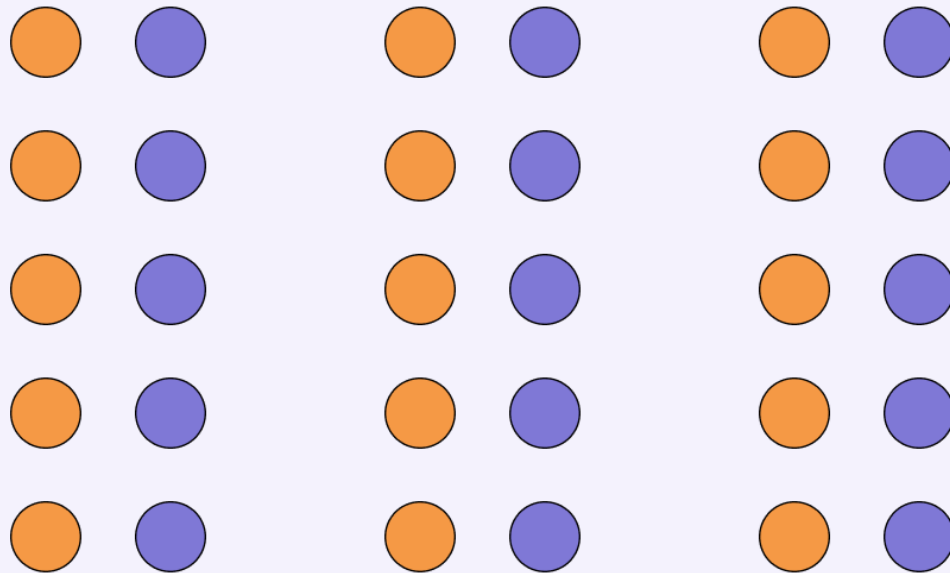
Gestalt Psychology

- We perceive objects in their entirety before their individual parts.



Source:<https://timesofindia.indiatimes.com/life-style/health-fitness/de-stress/optical-illusion-what-you-see-first-reveals-the-strongest-traits-of-your-personality/photostory/99866692.cms>

Proximity principle



<https://maze.co/blog/gestalt-principles/>

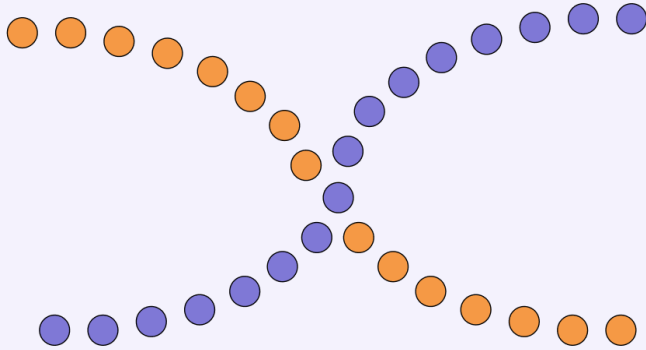
Gestalt Psychology

- **Figure-Ground** – We perceive objects (figures) as distinct from their background (ground). For example, in an image, we tend to focus on the subject while ignoring the backdrop.
- **Proximity** – Elements that are close together are perceived as part of the same group. For instance, dots placed near each other are seen as clusters rather than individual dots.
- **Similarity** – Objects that look alike (in shape, color, or size) are perceived as part of the same pattern or group.

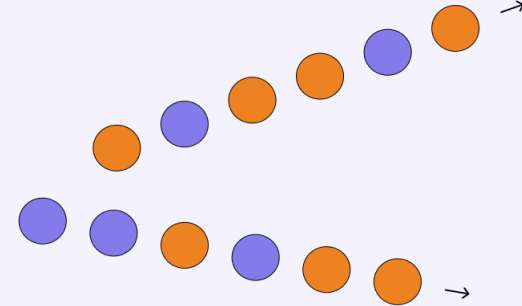
Gestalt Psychology

- **Closure** – Our brain tends to fill in gaps in incomplete images to perceive a complete, familiar shape. This explains why we recognize an incomplete circle or triangle.
- **Continuity** – The human eye prefers to see smooth, continuous lines and patterns rather than abrupt changes or disruptions.
- **Common Fate** – Elements that move in the same direction or have the same trajectory are perceived as a group.
- **Symmetry & Order (Prägnanz)** – We tend to organize visual elements in the simplest, most symmetrical, and balanced way.

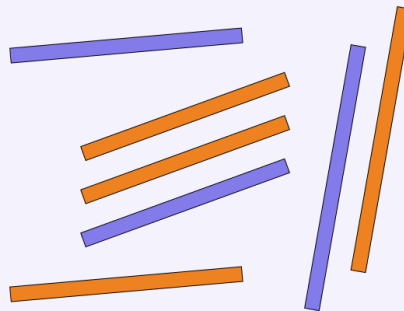
Continuity principle



Common fate principle



Parallelism principle



Segmentation as Clustering

- When viewed from a **clustering perspective**, segmentation involves grouping similar data points (or pixels in an image) based on shared characteristics
- Visual similarity can be based on: Brightness, Color, Position, Depth, Motion, etc

Segmentation as Clustering

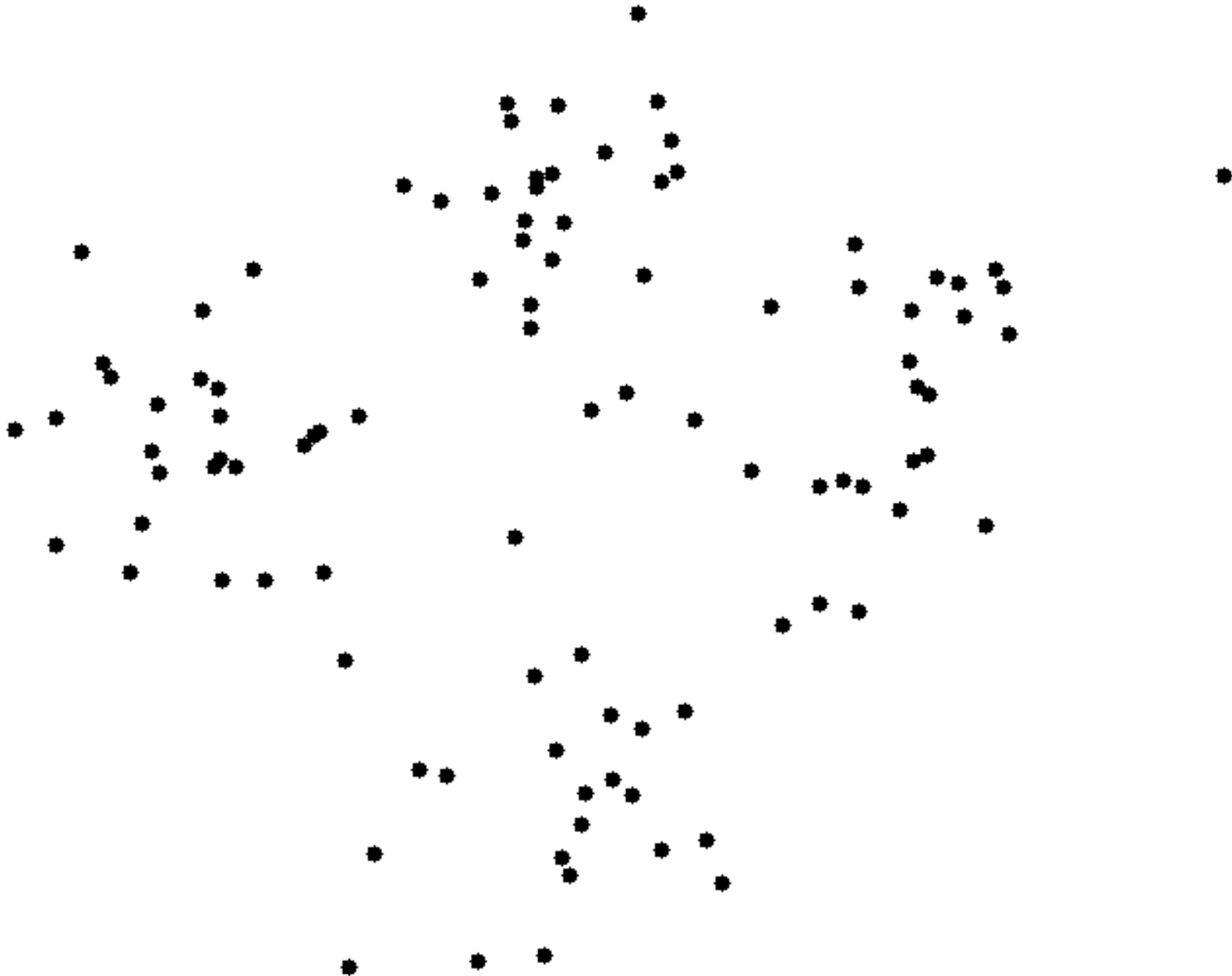
- Similarity measure:
 - Euclidean Distance (L2 Norm)
Measures straight-line distance between two points in an n -dimensional space:

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

K-Means Clustering

- K-Means partitions the data into **K clusters** by minimizing the sum of squared distances between points and their cluster centroids.
- **Algorithm Steps:**
 1. **Initialize:** Randomly select K initial centroids from the dataset.
 2. **Assignment Step:** Assign each data point to the nearest centroid (using a similarity measure, typically **Euclidean distance**).
 3. **Update Step:** Compute new centroids by averaging the points in each cluster.
 4. **Repeat:** Iterate steps 2 and 3 until convergence (i.e., centroids stop changing significantly).
 5. **Final Clusters:** The dataset is partitioned into K clusters.

K-Means Clustering



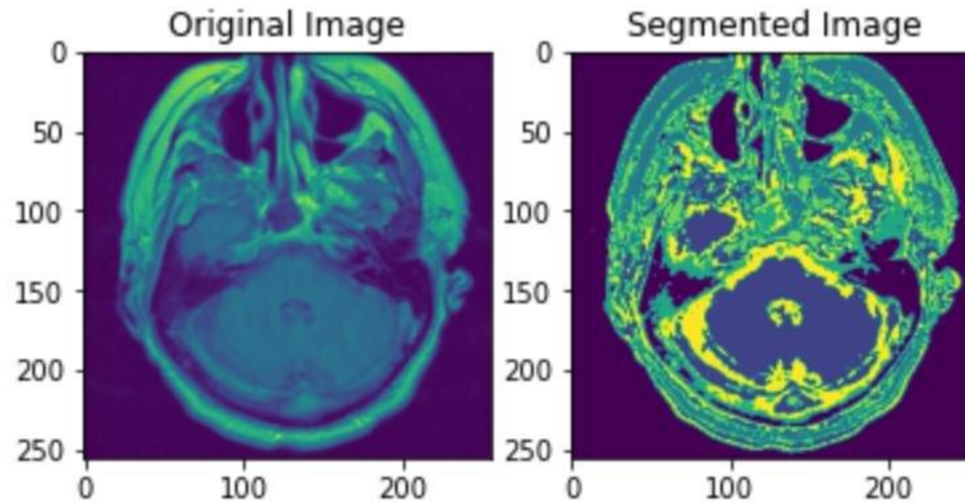
K-Means Clustering

- Mathematical Formulation:
 - Given a dataset $X = \{x_1, x_2, \dots, x_n\}$, K-Means aims to minimize the following objective function:

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

- where:
 - K is the number of clusters,
 - C_i represents the set of points in cluster i ,
 - μ_i is the centroid of cluster i ,
 - $\|x - \mu_i\|^2$ is the squared Euclidean distance.

K-Means Clustering



Shukla, Sameer. (2023). Enhancing Healthcare Insights, Exploring Diverse Use-Cases with K-means Clustering. International Journal of Management IT and Engineering. 13. 60-68.

K-Means Clustering

- **Advantages**

- Simple, efficient, and easy to implement.
- Scales well to large datasets.
- Works well when clusters are spherical and well-separated.

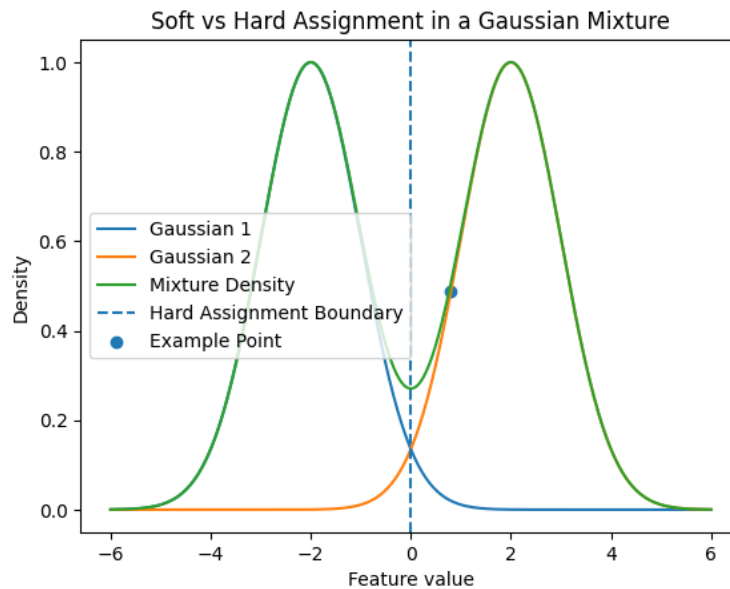
- **Limitations**

- Sensitive to **initialization** (randomly chosen centroids can lead to different results).
- Assumes **spherical clusters** (fails when clusters have irregular shapes).
- Does not handle **outliers** well.
- Requires **predefined K** (choosing the right number of clusters is non-trivial).

Mean-Shift Segmentation

- Mean Shift is a **non-parametric, density-based clustering algorithm** that is widely used for **image segmentation** and other clustering tasks.
- The algorithm iteratively **moves data points** towards regions of higher density (local maxima) in the feature space

Mean-Shift Segmentation



Mean-Shift Segmentation

- **Algorithm Steps:**

- 1.Initialize** each data point as a candidate cluster center.
- 2.Compute the mean of points** within a given bandwidth (radius) using a kernel function (commonly a Gaussian kernel).
- 3.Move the candidate center** to the computed mean.
- 4.Repeat until convergence**, i.e., when cluster centers no longer change significantly.
- 5.Merge similar clusters** if they are within the same bandwidth.

Mean-Shift Segmentation

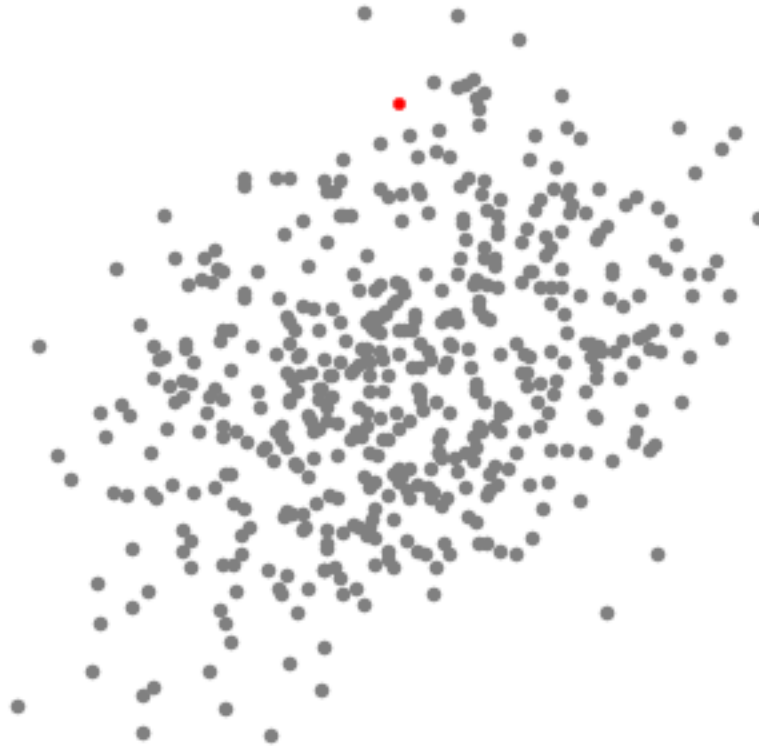
- Mathematical Formulation:
 - At each iteration, the mean shift update is:

$$m(x) = \frac{\sum_i K(x_i - x)x_i}{\sum_i K(x_i - x)}$$

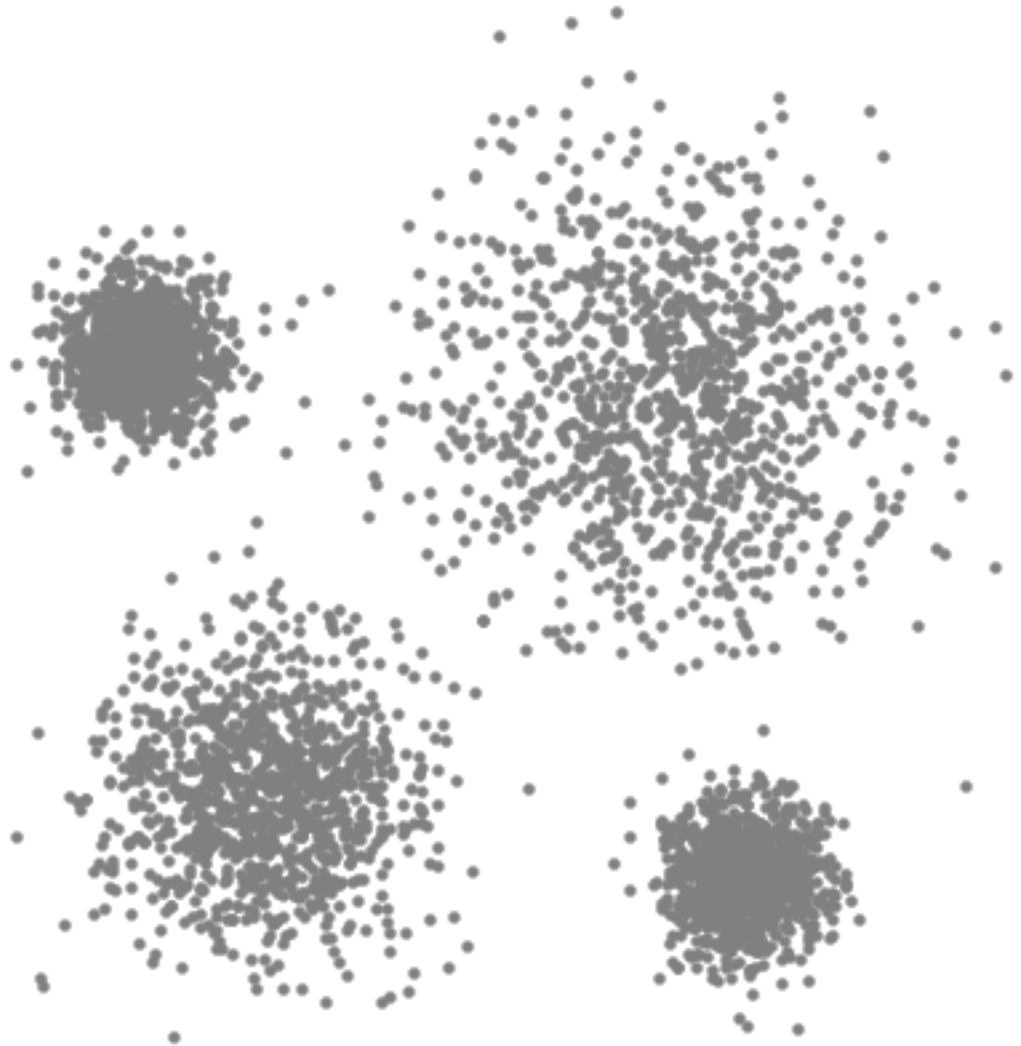
– where:

- x is the current point,
- x_i are neighboring points within the bandwidth,
- $K(x_i - x)$ is a kernel function (usually Gaussian).

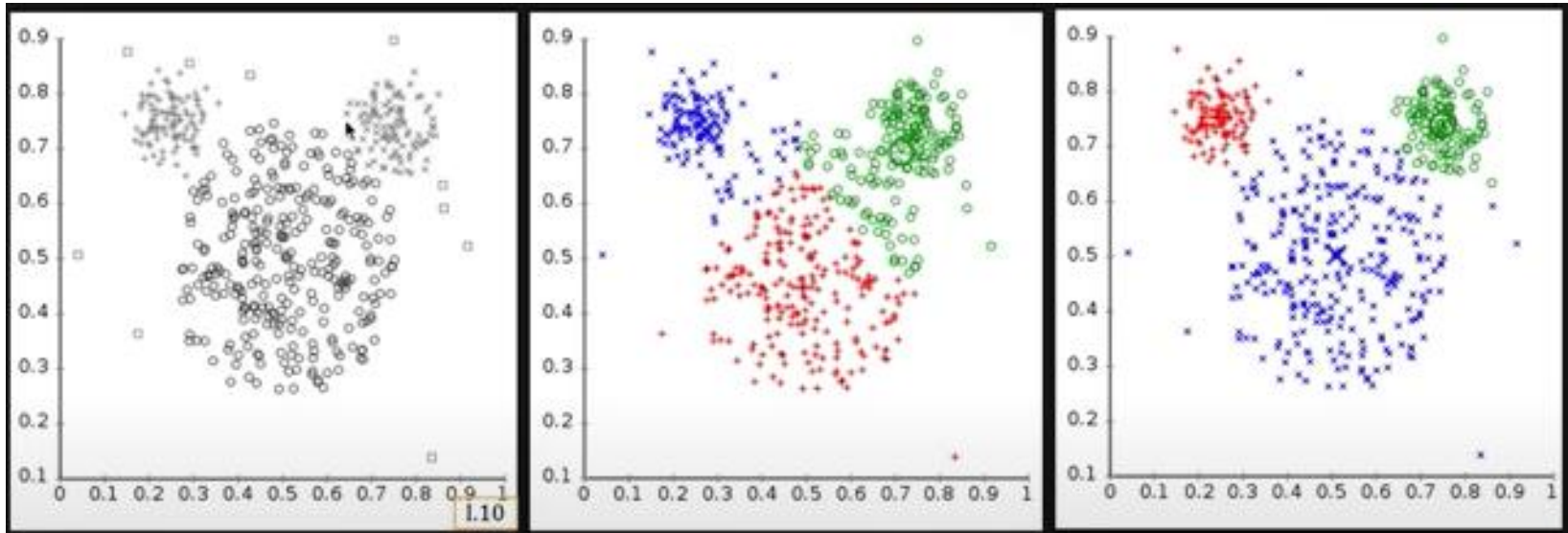
Mean-Shift Segmentation



Mean-Shift Segmentation



Mean-Shift Segmentation



Original data

K-Means

Mean Shift

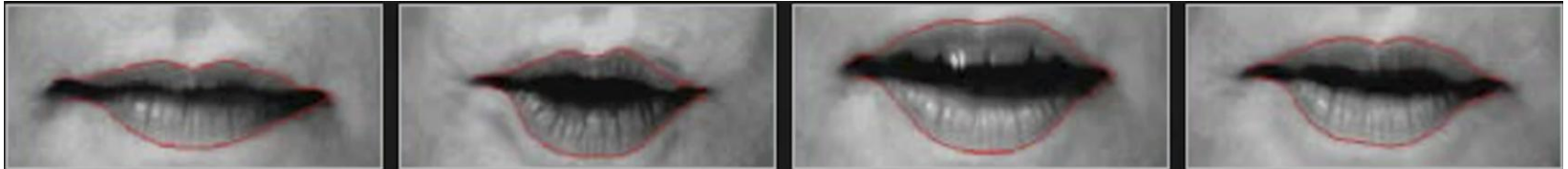
Mean-Shift Segmentation



Active Contour

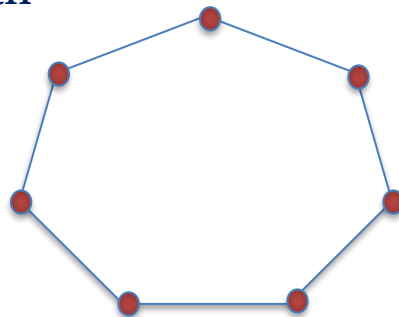
- **Initial Input:** A rough contour outlining the object.
- **Objective:** Adjust the contour to match the object's true boundary.
- **Active Contour Model:**
 - Progressively modifies the contour shape.
 - Moves towards regions with strong edges (high gradient pixels).
 - Maintains a smooth shape during adjustment.
- **Also referred to as "Snakes."**

Active Contour



Represing a contour

- **Contour $\mathbf{v}$:** An ordered list of 2D vertices (control points) connected by straight lines of fixed length



$$\mathbf{v} = \{v_i = (x_i, y_i) \mid i = 0, 1, 2, \dots, n - 1\}$$





Contour Deformation



Image with Initial
Contour

Gradient Magnitude

Blurred Gradient

- Maximize Sum of Gradient Magnitude Square

$$\equiv \text{Minimize } E_{image} = - \sum_{i=0}^{n-1} \|\nabla n_{\sigma} * I(v_i)\|^2$$

Contour Deformation

The total energy function is given by:

$$E_{\text{snake}} = E_{\text{internal}} + E_{\text{image}} + E_{\text{external}}$$

where:

Internal Energy (Smoothness and Elasticity)

$$E_{\text{internal}} = \sum_{i=0}^{n-1} [\alpha |v_{i+1} - v_i|^2 + \beta |v_{i+1} - 2v_i + v_{i-1}|^2]$$

First term controls elasticity (length preservation).

Second term controls curvature (smoothness).

α, β are weights controlling these properties.

Contour Deformation

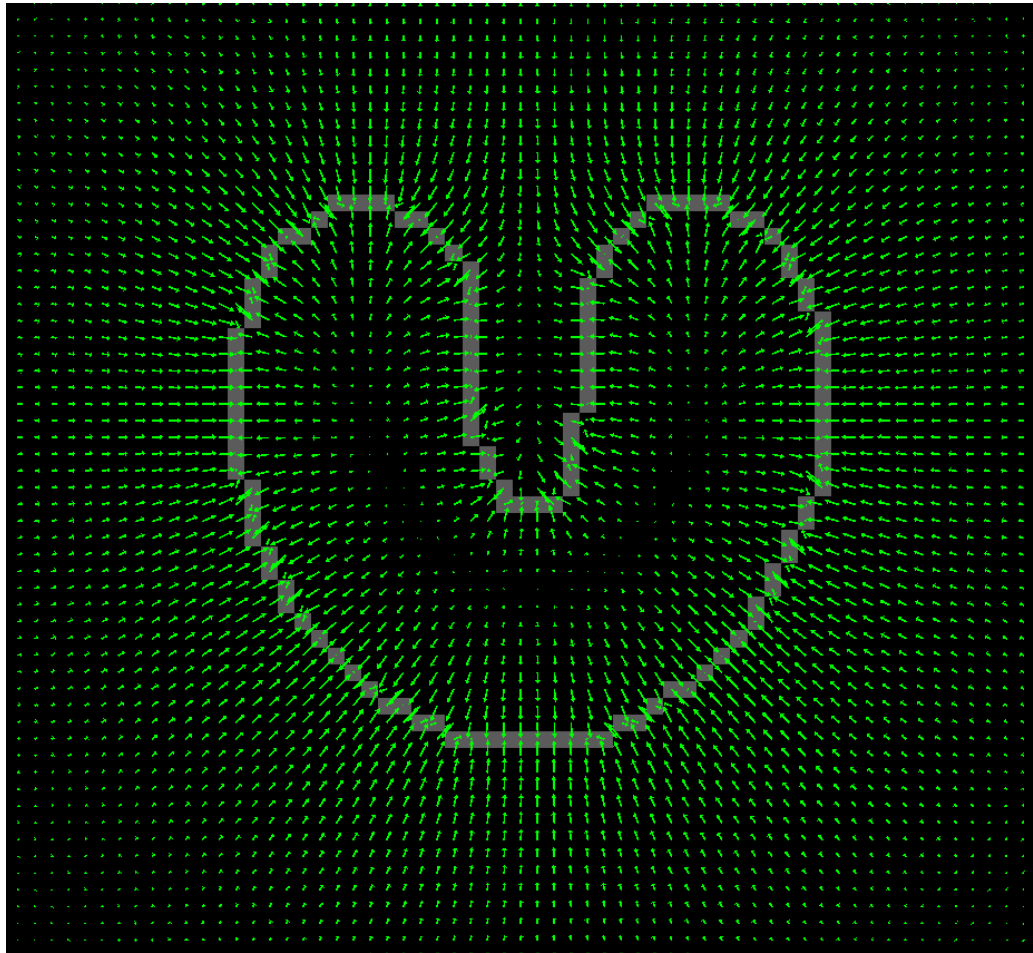
External Energy (Balloon Force, Constraints)

$$E_{\text{external}} = \sum_{i=0}^{n-1} \gamma F(v_i)$$

$F(v_i)$ is an external force, such as a balloon force (expanding contour).

γ is a weight controlling its effect.

GVF



<https://iacl.ece.jhu.edu/Projects/gvf/gvf.gif>

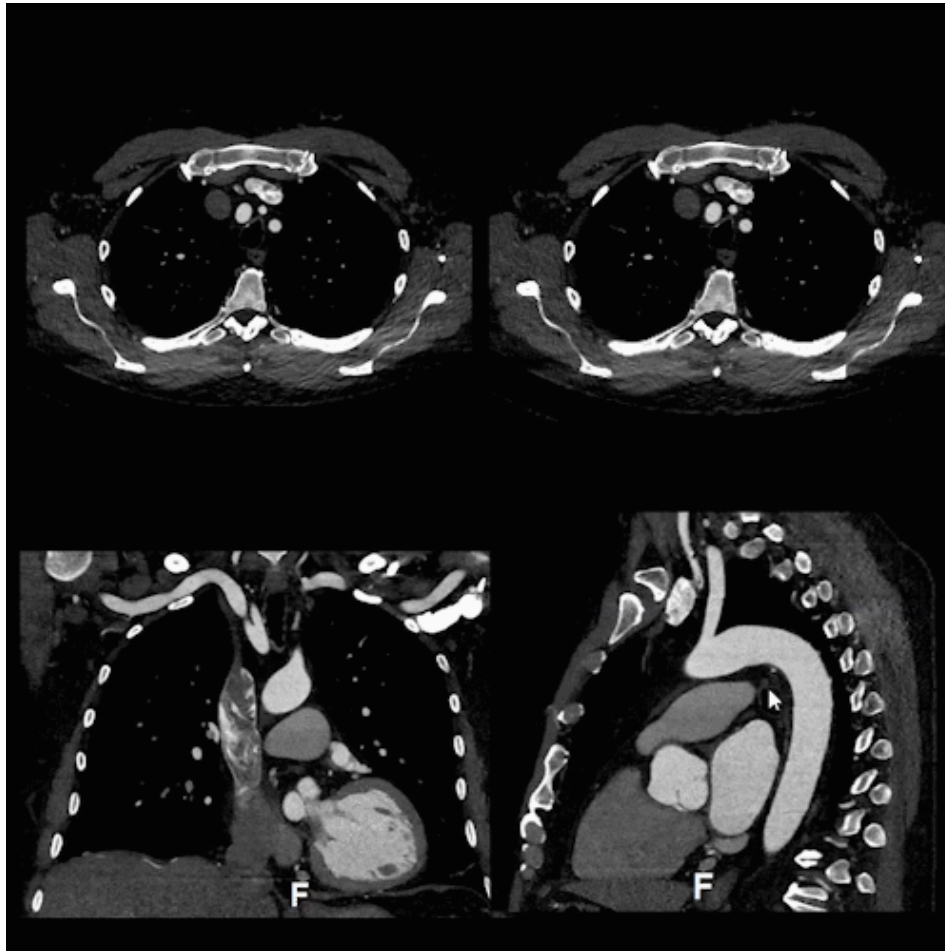
Region-Based Segmentation

- Region-Growing:
 - Group pixels into homogeneous regions based on similarity.
 - Starts from **seed pixels** and **grows** by adding neighboring pixels **with similar properties**.

Region-Based Segmentation

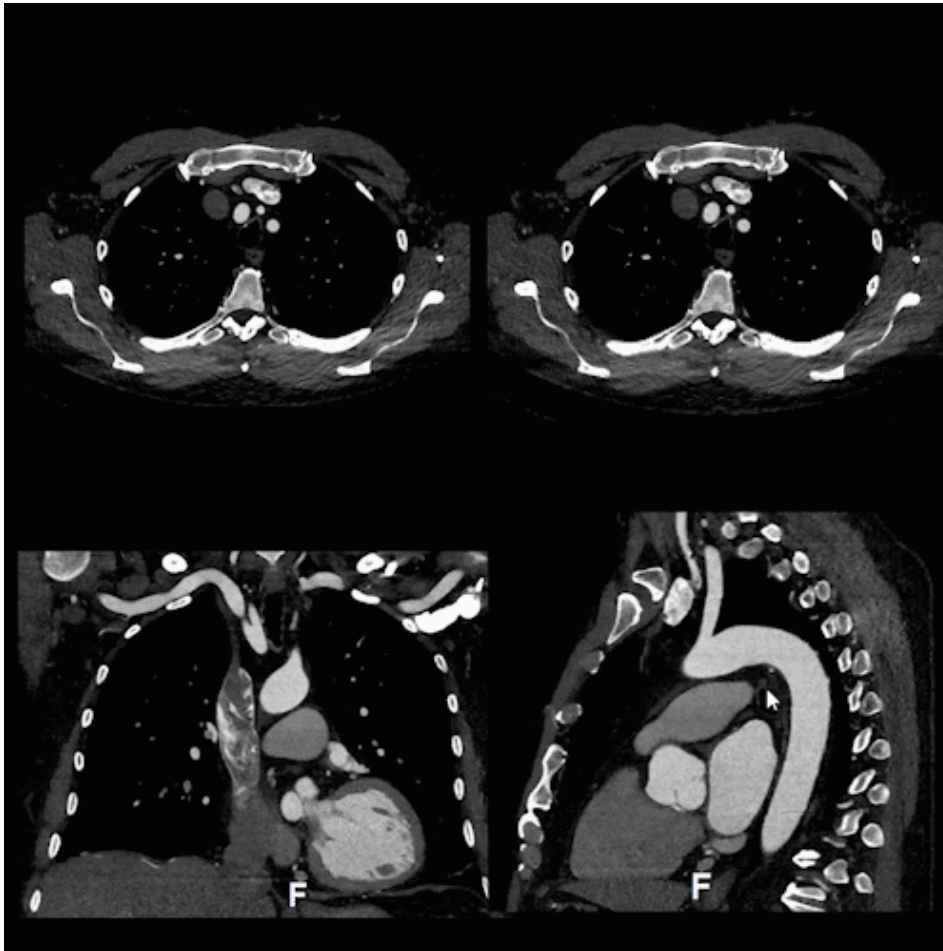
- Region-Growing:
 - Starts with **one or more seed pixels**.
 - **Expands by adding neighboring pixels** if they meet a **similarity condition**.
 - One needs to define the similarity criteria:
 - **Intensity** (e.g., grayscale value difference)
 - **Color** (e.g., RGB/HSV difference)
 - **Texture** (e.g., local variance)
 - **Statistical measures** (e.g., mean, variance)
 - Stops when **no more pixels** satisfy the condition.

Region-Based Segmentation



Region-Based Segmentation

Region Growing



Region-Based: Split and merge segmentation

- Define the criterion to be used for homogeneity
- Split the image into equal size regions
- Calculate homogeneity for each region
- If the region is homogeneous, then merge it with neighbors
- The process is repeated until all regions pass the homogeneity test

Split and merge segmentation

